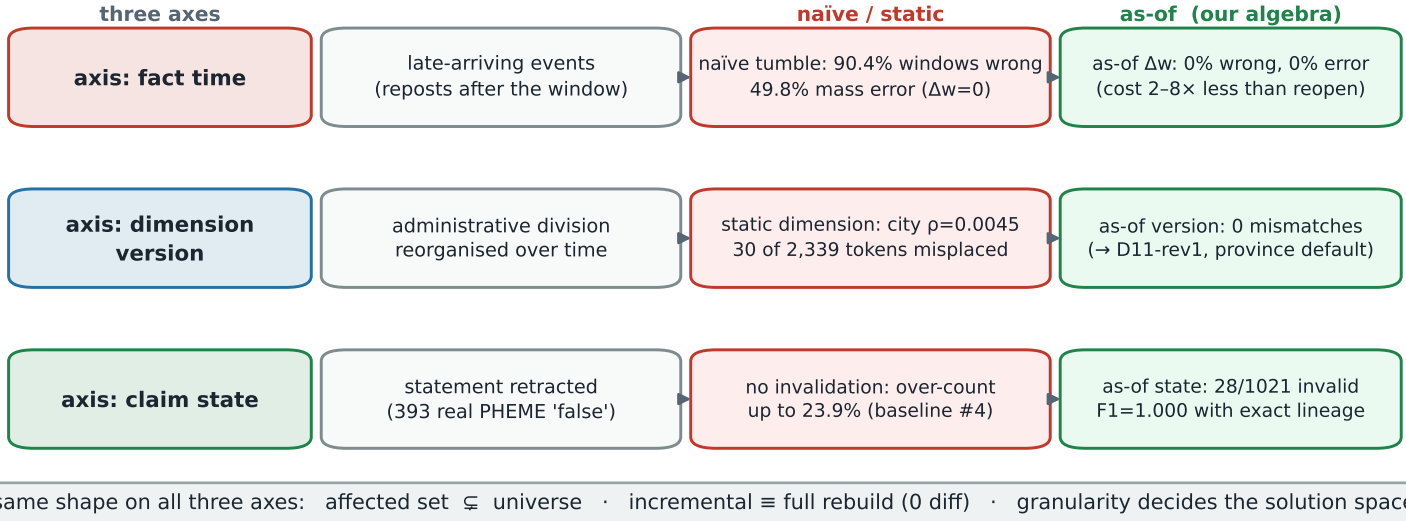
three axes
naïve / static
as-of (our algebra)
axis: fact time
late-arriving events
(reposts after the window)
naïve tumble: 90.4% windows wrong
49.8% mass error (Δw=0)
as-of Δw: 0% wrong, 0% error
(cost 2–8× less than reopen)
axis: dimension
version
administrative division
reorganised over time
static dimension: city ρ=0.0045
30 of 2,339 tokens misplaced
as-of version: 0 mismatches
(→ D11-rev1, province default)
axis: claim state
statement retracted
(393 real PHEME 'false')
no invalidation: over-count
up to 23.9% (baseline #4)
as-of state: 28/1021 invalid
F1=1.000 with exact lineage
same shape on all three axes: affected set ⊊ universe · incremental ≡ full rebuild (0 diff) · granularity decides the solution space